Solving Perfect Multicollinearity with Nested Geographic Variables in R

Paul Mosquera

Contents

1 The Problem: Nested Geographic Variables in Regression **2**
1.1 The Mathematical Reason . . . 2
1.2 The Dataset . . . 2
1.3 The Hierarchical Structure . . . 3
1.4 Demonstrating the Failure . . . 4

2 Diagnosis: Confirming the Problem **4**
2.1 Variance Inflation Factor (VIF) . . . 4
2.2 Rank of the Design Matrix . . . 4

3 Statistical Assumptions Check **5**
3.1 Normality of the Response Variable . . . 5
3.2 Homoscedasticity Across Geographic Groups . . . 7

4 The Solution **10**
4.1 Step 1: Choosing the BASE Province . . . 10
4.2 Why Does This Work? . . . 11
4.3 Step 2: Applying Sum-to-Zero Contrasts . . . 12
4.4 Fitting the Model . . . 12

5 Verification: Did the Solution Work? **13**
5.1 VIF After the Fix . . . 13
5.2 Rank of the New Design Matrix . . . 14
5.3 Visual Comparison of VIF . . . 14
5.4 Model Diagnostics . . . 15

6 Interpretation: Reading Coefficients Under Sum-to-Zero Contrasts **17**
6.1 The Intercept . . . 17
6.2 Province Coefficients . . . 17
6.3 A Concrete Example . . . 19

7 Applicability and Limitations **19**
7.1 When This Solution Is Appropriate . . . 19
7.2 When to Consider Alternatives . . . 19
7.3 The General Recipe . . . 19
7.4 Summary . . . 19

1 The Problem: Nested Geographic Variables in Regression

Imagine you are building a model to explain housing prices across municipalities in Belgium. Your dataset contains two geographic variables: the administrative region each municipality belongs to (AdminArea), and its province (Province). Naturally, you want to include both because regions capture broad geographic effects and provinces capture finer local differences.

The problem is that these two variables are not independent. Every province belongs to exactly one region, always. Hainaut is always in Waals_Gewest. Limburg is always in Vlaams_Gewest. There are no exceptions. This rigid hierarchical structure means that once you know the province of a municipality, you already know its region with 100% certainty.

When you try to include both variables in a regression model, R cannot distinguish the effect of the region from the effect of the province. The model literally cannot be estimated. This is called perfect multicollinearity.

1.1 The Mathematical Reason

In linear regression, the coefficients are estimated using the formula:

$$\hat{\beta} = (X'X)^{-1}X'y$$

where X is the design matrix containing all predictor variables. For this formula to work, the matrix $X'X$ must be invertible, which requires that no column of X can be written as an exact linear combination of other columns.

When Province is nested within AdminArea, the region column in X is completely determined by the province columns:

$$\det(X'X) = 0 \implies (X'X)^{-1} \text{ does not exist}$$

If you sum the province dummy columns belonging to Vlaams_Gewest, you get exactly the AdminArea dummy column:

$$x_{\text{Antwerpen}} + x_{\text{Limburg}} + x_{\text{Oost_Vl}} + \cdots = x_{\text{Vlaams_Gewest}}$$

1.2 The Dataset

We use housing price data from Belgium with 566 municipalities, focusing on two administrative regions: Vlaams_Gewest (Flemish Region) and Waals_Gewest (Walloon Region). Each municipality belongs to exactly one province, which belongs to exactly one region.

```r
path <- paste0("C:/Users/paulm/Desktop/Projects/", "nested-geographic-variables-multicollinearity/",
    "data/HousePrices.txt")
df <- read.table(path, header = TRUE)

df <- df |>
    filter(AdminArea %in% c("Vlaams_Gewest", "Waals_Gewest"))

df$AdminArea <- factor(df$AdminArea, levels = c("Vlaams_Gewest",
    "Waals_Gewest"))
df$Province <- droplevels(factor(df$Province))

cat("Rows after filter:", nrow(df), "\n")
```

```
## Rows after filter: 566
```

```r
cat("AdminArea levels:", levels(df$AdminArea), "\n")
```

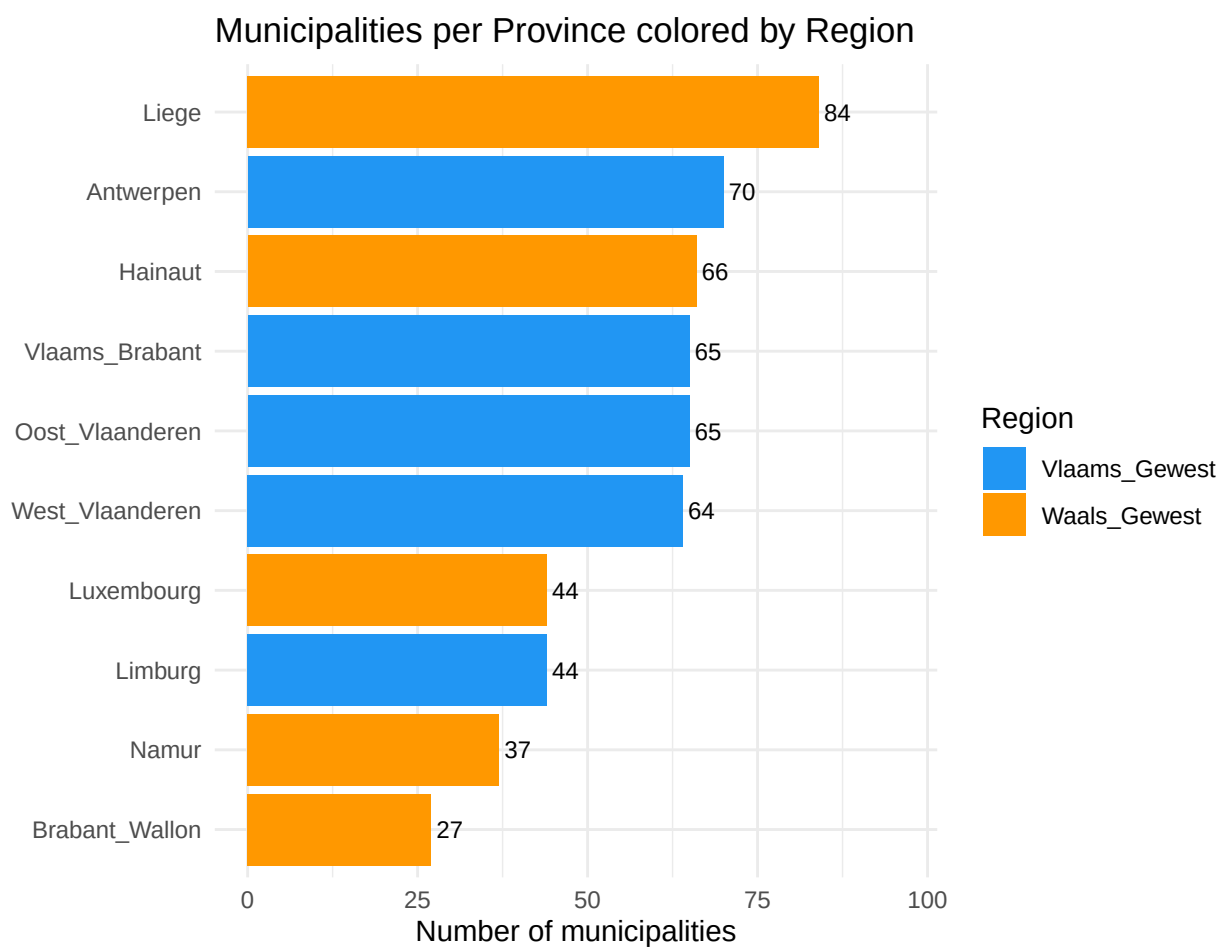
```
## AdminArea levels: Vlaams_Gewest Waals_Gewest
```

```r
cat("Province levels:", levels(df$Province), "\n")
```

```
## Province levels: Antwerpen Brabant_Wallon Hainaut Liege Limburg Luxembourg Namur Oost_Vlaanderen Vla
```

1.3 The Hierarchical Structure

```r
hierarchy <- df |>
    count(AdminArea, Province) |>
    arrange(AdminArea, desc(n))

ggplot(hierarchy, aes(x = reorder(Province, n), y = n, fill = AdminArea)) +
    geom_col() + geom_text(aes(label = n), hjust = -0.2,
    size = 3) + coord_flip() + scale_fill_manual(values = c(Vlaams_Gewest = "#2196F3",
    Waals_Gewest = "#FF9800")) + labs(title = "Municipalities per Province colored by Region",
    x = NULL, y = "Number of municipalities", fill = "Region") +
    theme_minimal(base_size = 11) + expand_limits(y = max(hierarchy$n) *
    1.15)
```



The color pattern is the key insight: every province bar is a single color. This perfect separation is precisely

what causes the model to fail.

1.4 Demonstrating the Failure

```r
model_naive <- lm(med_price_house ~ AdminArea + Province,
    data = df)
summary(model_naive)
```

```
##
## Call:
## lm(formula = med_price_house ~ AdminArea + Province, data = df)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -99.755 -20.920  -3.104  19.381 163.409
##
## Coefficients: (1 not defined because of singularities)
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)               240.585      4.068  59.139  < 2e-16 ***
## AdminAreaWaals_Gewest    -85.558      6.918 -12.367  < 2e-16 ***
## ProvinceBrabant_Wallon   101.564      8.615  11.789  < 2e-16 ***
## ProvinceHainaut          -20.372      6.990  -2.914 0.003708 **
## ProvinceLiege             17.596      6.716   2.620 0.009032 **
## ProvinceLimburg          -55.620      6.548  -8.494  < 2e-16 ***
## ProvinceLuxembourg        -4.107      7.592  -0.541 0.588793
## ProvinceNamur                 NA         NA      NA       NA
## ProvinceOost_Vlaanderen  -21.225      5.863  -3.620 0.000321 ***
## ProvinceVlaams_Brabant    13.784      5.863   2.351 0.019070 *
## ProvinceWest_Vlaanderen  -38.830      5.887  -6.596 9.82e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 34.04 on 556 degrees of freedom
## Multiple R-squared:  0.5955, Adjusted R-squared:  0.5889
## F-statistic: 90.94 on 9 and 556 DF,  p-value: < 2.2e-16
```

Several province coefficients appear as NA. This is R silently telling you that it cannot estimate those coefficients because the information they would carry is perfectly confounded by the AdminArea variable.

2 Diagnosis: Confirming the Problem

2.1 Variance Inflation Factor (VIF)

$$VIF_j = \frac{1}{1 - R_j^2}$$

```r
tryCatch(car::vif(model_naive), error = function(e) cat("VIF Error:",
    conditionMessage(e), "\n"))
```

```
## VIF Error: there are aliased coefficients in the model
```

2.2 Rank of the Design Matrix

```r
X_naive <- model.matrix(~AdminArea + Province, data = df)
cat("Number of columns in X:", ncol(X_naive), "\n")
```

```
## Number of columns in X: 11
```

```r
cat("Rank of X:", qr(X_naive)$rank, "\n")
```

```
## Rank of X: 10
```

```r
cat("Rank deficiency:", ncol(X_naive) - qr(X_naive)$rank,
    "\n")
```

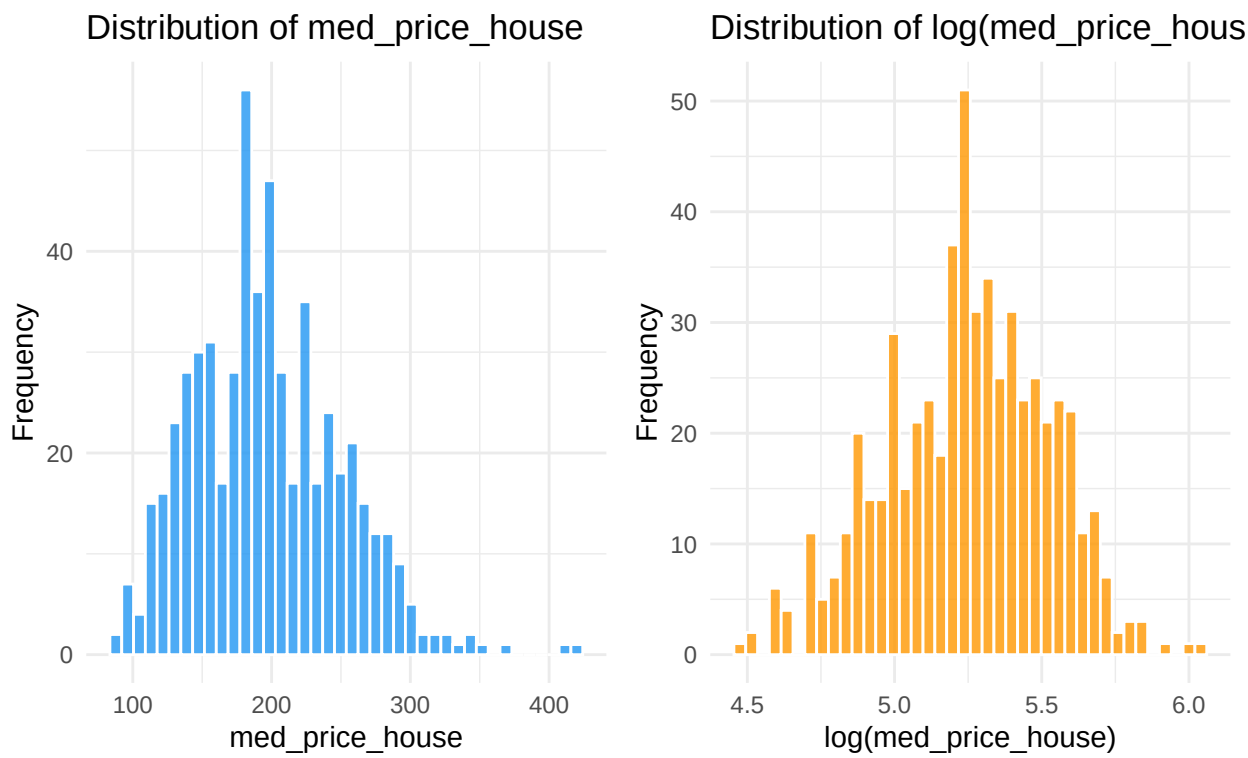
```
## Rank deficiency: 1
```

3 Statistical Assumptions Check

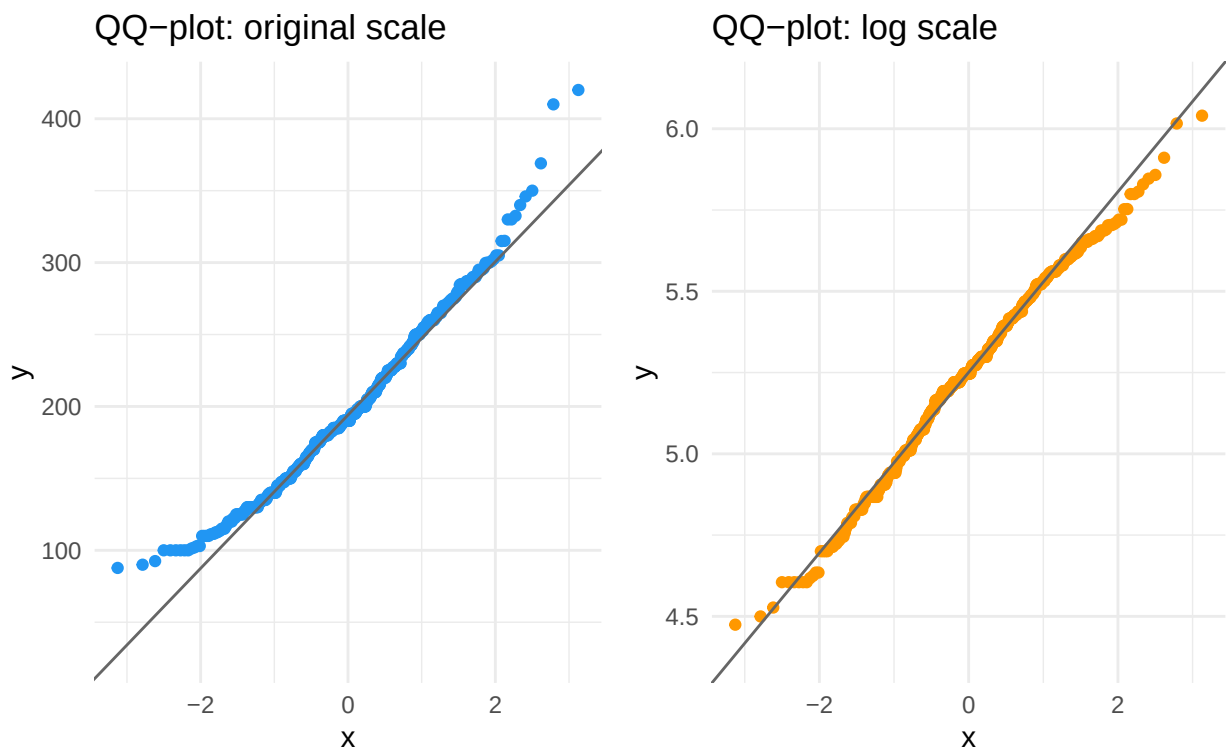
Before applying the solution, we verify whether the conditions for classical linear regression are reasonably met. The goal here is not to validate a final predictive model but to establish whether this methodological solution is applicable in this context. We are using a deliberately simple model to demonstrate the solution clearly, not to build the most accurate model possible.

3.1 Normality of the Response Variable

```r
df$log_price <- log(df$med_price_house)

p1 <- ggplot(df, aes(x = med_price_house)) + geom_histogram(bins = 40,
    fill = "#2196F3", color = "white", alpha = 0.8) + labs(title = "Distribution of med_price_house",
    x = "med_price_house", y = "Frequency") + theme_minimal(base_size = 11)

p2 <- ggplot(df, aes(x = log_price)) + geom_histogram(bins = 40,
    fill = "#FF9800", color = "white", alpha = 0.8) + labs(title = "Distribution of log(med_price_house
    x = "log(med_price_house)", y = "Frequency") + theme_minimal(base_size = 11)

gridExtra::grid.arrange(p1, p2, ncol = 2)
```




```r
p_qq <- ggplot(df, aes(sample = med_price_house)) + stat_qq(color = "#2196F3") +
    stat_qq_line(color = "gray40") + labs(title = "QQ-plot: original scale") +
    theme_minimal(base_size = 11)

p_qq_log <- ggplot(df, aes(sample = log_price)) + stat_qq(color = "#FF9800") +
    stat_qq_line(color = "gray40") + labs(title = "QQ-plot: log scale") +
    theme_minimal(base_size = 11)

gridExtra::grid.arrange(p_qq, p_qq_log, ncol = 2)
```



The original scale shows moderate right skewness. The log transformation improves symmetry noticeably. Despite this, we keep the response in its original scale for a deliberate reason: the objective is to demonstrate the methodological solution with directly interpretable coefficients in monetary units. With a sample of 566 observations, the Central Limit Theorem provides sufficient protection for valid OLS inference even with moderate skewness.

3.2 Homoscedasticity Across Geographic Groups

This is the most critical assumption for our specific solution. We use Levene's test, which is robust to departures from normality.

```
cat("Levene's test across AdminArea:\n")
```

```
## Levene's test across AdminArea:
```

```
leveneTest(med_price_house ~ AdminArea, data = df)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value Pr(>F)
## group   1  0.0038 0.9508
##       564
```

```
cat("Levene's test across Province:\n")
```

```
## Levene's test across Province:
```

```
leveneTest(med_price_house ~ Province, data = df)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value    Pr(>F)
## group   9  4.8126 3.328e-06 ***
##       556
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
var_by_region <- df |>
    group_by(AdminArea) |>
    summarise(n = n(), mean = round(mean(med_price_house),
        2), sd = round(sd(med_price_house), 2), variance = round(var(med_price_house),
        2), .groups = "drop")

knitr::kable(var_by_region, caption = "Variance of housing prices by AdminArea",
    booktabs = TRUE)
```

Table 1: Variance of housing prices by AdminArea

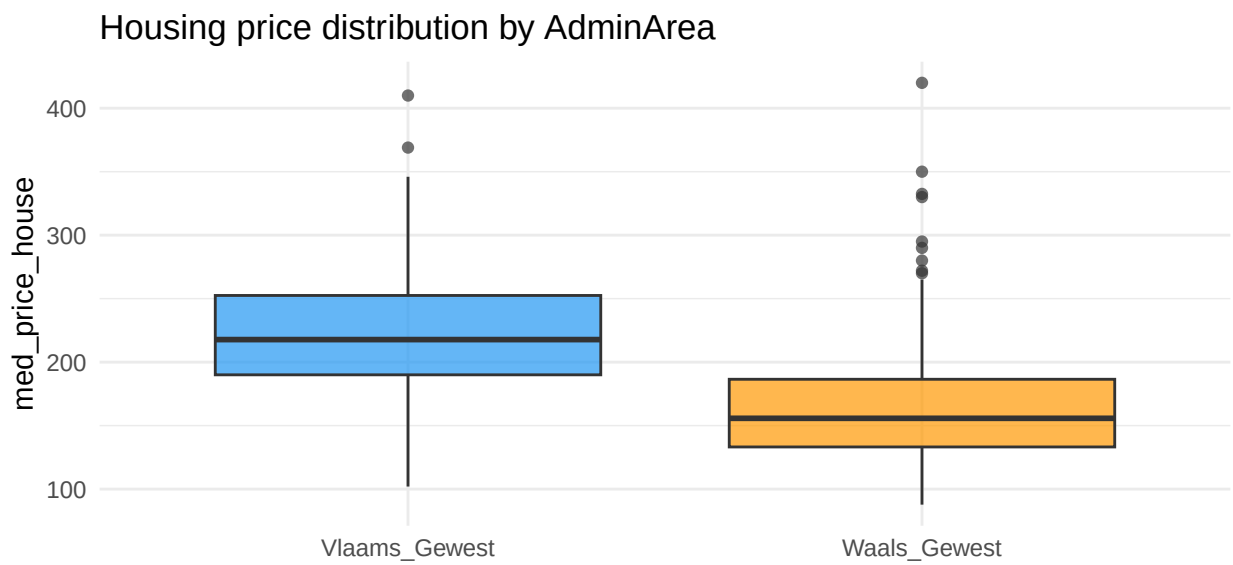
AdminArea	n	mean	sd	variance
Vlaams_Gewest	308	223.00	42.67	1821.06
Waals_Gewest	258	165.47	47.03	2212.27

```r
var_by_province <- df |>
    group_by(AdminArea, Province) |>
    summarise(n = n(), mean = round(mean(med_price_house),
        2), sd = round(sd(med_price_house), 2), variance = round(var(med_price_house),
        2), .groups = "drop") |>
    arrange(AdminArea, desc(variance))

knitr::kable(var_by_province, caption = "Variance of housing prices by Province",
    booktabs = TRUE)
```

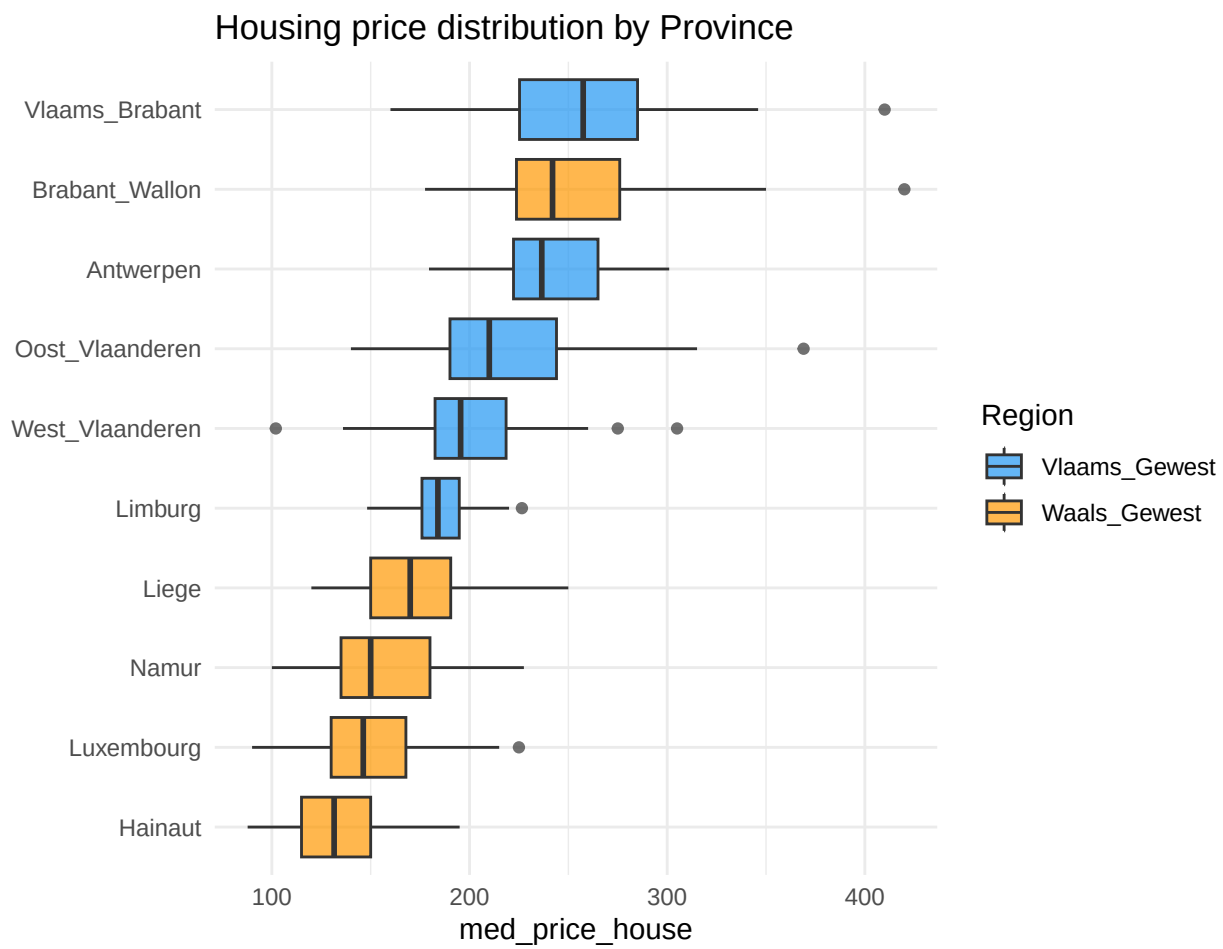
Table 2: Variance of housing prices by Province

AdminArea	Province	n	mean	sd	variance
Vlaams_Gewest	Vlaams_Brabant	65	254.37	46.58	2169.30
Vlaams_Gewest	Oost_Vlaanderen	65	219.36	38.41	1475.49
Vlaams_Gewest	West_Vlaanderen	64	201.75	33.16	1099.71
Vlaams_Gewest	Antwerpen	70	240.58	30.51	930.69
Vlaams_Gewest	Limburg	44	184.96	17.58	309.18
Waals_Gewest	Brabant_Wallon	27	256.59	54.60	2981.70
Waals_Gewest	Namur	37	155.03	33.11	1096.59
Waals_Gewest	Liege	84	172.62	29.65	878.94
Waals_Gewest	Luxembourg	44	150.92	27.80	772.92
Waals_Gewest	Hainaut	66	134.66	25.07	628.72

```r
ggplot(df, aes(x = AdminArea, y = med_price_house, fill = AdminArea)) +
    geom_boxplot(alpha = 0.7) + scale_fill_manual(values = c(Vlaams_Gewest = "#2196F3",
    Waals_Gewest = "#FF9800")) + labs(title = "Housing price distribution by AdminArea",
    x = NULL, y = "med_price_house") + theme_minimal(base_size = 11) +
    theme(legend.position = "none")
```




```r
ggplot(df, aes(x = reorder(Province, med_price_house, FUN = median),
    y = med_price_house, fill = AdminArea)) + geom_boxplot(alpha = 0.7) +
    coord_flip() + scale_fill_manual(values = c(Vlaams_Gewest = "#2196F3",
    Waals_Gewest = "#FF9800")) + labs(title = "Housing price distribution by Province",
    x = NULL, y = "med_price_house", fill = "Region") +
    theme_minimal(base_size = 11)
```



If Levene's test rejects homoscedasticity, this does not invalidate the solution. OLS estimators remain unbiased under heteroscedasticity. What changes is that standard errors may be slightly underestimated, which affects the precision of hypothesis tests. For a methodological demonstration like this one, that is an acceptable limitation. If formal inference were the primary goal, heteroscedasticity-consistent standard errors or a multilevel model with `lme4` would be more appropriate.

4 The Solution

The fix has two steps. First, we create a BASE reference category by merging one representative province from each region. Second, we apply sum-to-zero contrasts to both AdminArea and the recoded province variable.

4.1 Step 1: Choosing the BASE Province

```
base_provinces <- df |>
    count(AdminArea, Province) |>
    group_by(AdminArea) |>
    slice_max(n, n = 1) |>
    ungroup()

knitr::kable(base_provinces, caption = "Selected BASE province per region",
    booktabs = TRUE)
```

Table 3: Selected BASE province per region

AdminArea	Province	n
Vlaams_Gewest	Antwerpen	70
Waals_Gewest	Liege	84

```
df$Province2 <- as.character(df$Province)

for (i in seq_len(nrow(base_provinces))) {
    area <- base_provinces$AdminArea[i]
    prov <- base_provinces$Province[i]
    df$Province2[df$AdminArea == area & df$Province == prov] <- "BASE"
}

lv <- c(setdiff(unique(df$Province2), "BASE"), "BASE")
df$Province2 <- factor(df$Province2, levels = lv)

df |>
    count(Province2) |>
    arrange(desc(n)) |>
    knitr::kable(caption = "Municipality count per Province2 after BASE recoding",
        booktabs = TRUE)
```

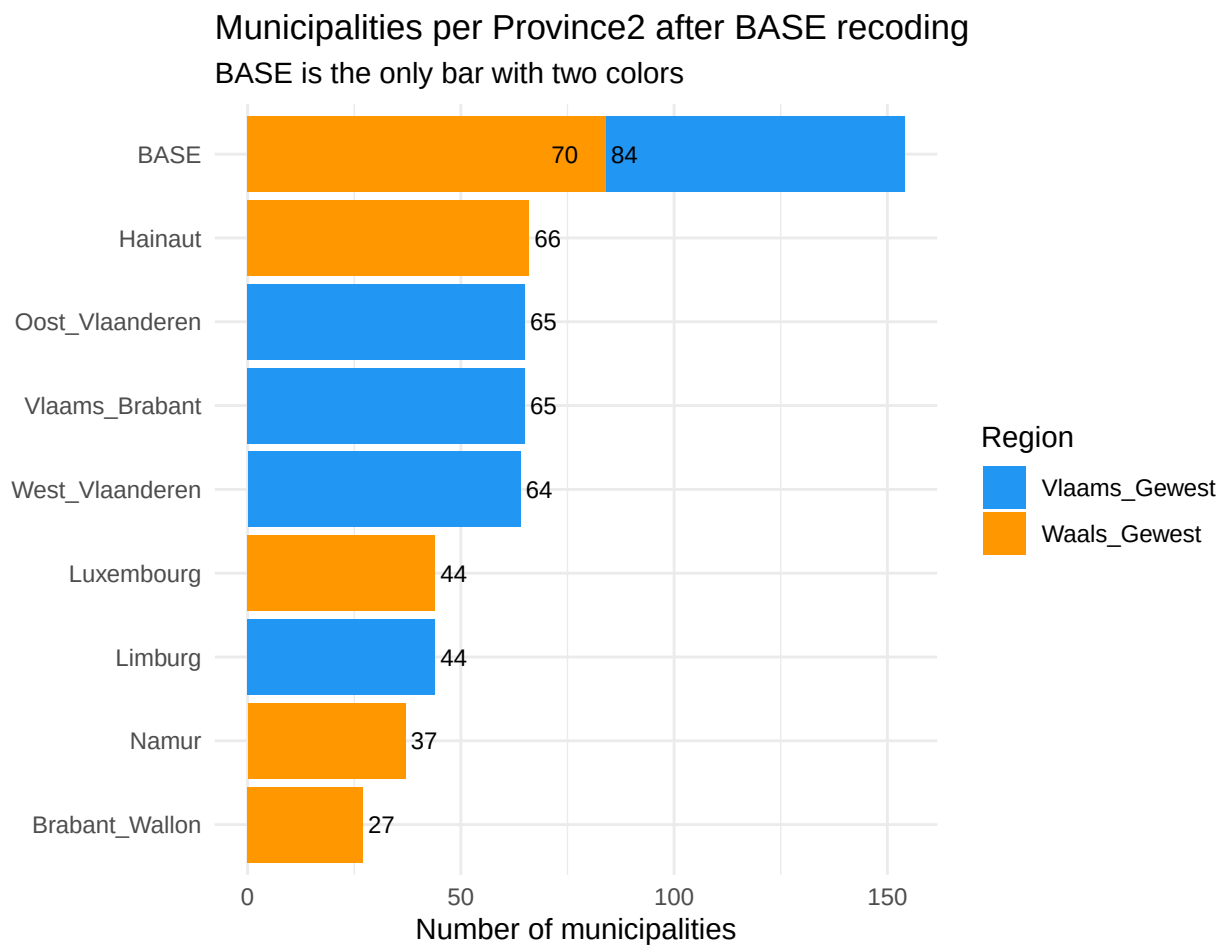
Table 4: Municipality count per Province2 after BASE recoding

Province2	n
BASE	154
Hainaut	66
Vlaams_Brabant	65
Oost_Vlaanderen	65
West_Vlaanderen	64
Limburg	44
Luxembourg	44
Namur	37
Brabant_Wallon	27

4.2 Why Does This Work?

$$\text{Province2}_i = \text{BASE} \not\Rightarrow \text{AdminArea}_i \text{ is determined}$$

```
hierarchy2 <- df |>
    count(AdminArea, Province2) |>
    arrange(AdminArea, desc(n))

ggplot(hierarchy2, aes(x = reorder(Province2, n), y = n,
    fill = AdminArea)) + geom_col() + geom_text(aes(label = n),
    hjust = -0.2, size = 3) + coord_flip() + scale_fill_manual(values = c(Vlaams_Gewest = "#2196F3",
    Waals_Gewest = "#FF9800")) + labs(title = "Municipalities per Province2 after BASE recoding",
    subtitle = "BASE is the only bar with two colors", x = NULL,
    y = "Number of municipalities", fill = "Region") + theme_minimal(base_size = 11) +
    expand_limits(y = max(hierarchy2$n) * 1.15)
```



4.3 Step 2: Applying Sum-to-Zero Contrasts

$$\sum_{k=1}^{K} \beta_k = 0$$

```
kA <- nlevels(df$AdminArea)
kP <- nlevels(df$Province2)

CA <- contr.sum(kA)
CP <- contr.sum(kP)

colnames(CA) <- levels(df$AdminArea)[1:(kA - 1)]
colnames(CP) <- levels(df$Province2)[1:(kP - 1)]

contrasts(df$AdminArea) <- CA
contrasts(df$Province2) <- CP
```

4.4 Fitting the Model

With the recoding and contrasts in place, we fit a deliberately simple model that includes both geographic levels simultaneously along with mean income as a continuous predictor. The simplicity is intentional: the goal is to demonstrate the methodological solution clearly, not to build an exhaustive predictive model.

```r
df$Mean_income_c <- df$Mean_income - mean(df$Mean_income,
    na.rm = TRUE)

model_solution <- lm(med_price_house ~ AdminArea + Province2 +
    Mean_income_c, data = df)

summary(model_solution)
```

```
##
## Call:
## lm(formula = med_price_house ~ AdminArea + Province2 + Mean_income_c,
##     data = df)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -72.828 -17.605  -0.276  14.715 105.490
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)              195.4839     1.1884 164.496  < 2e-16 ***
## AdminAreaVlaams_Gewest    24.2158     2.1045  11.506  < 2e-16 ***
## Province2Vlaams_Brabant   -1.6847     3.8649  -0.436  0.66308
## Province2West_Vlaanderen  -4.0536     3.8535  -1.052  0.29330
## Province2Oost_Vlaanderen -11.1386     3.6627  -3.041  0.00247 **
## Province2Limburg         -25.7090     4.2344  -6.071 2.35e-09 ***
## Province2Brabant_Wallon   52.2938     5.3002   9.866  < 2e-16 ***
## Province2Hainaut         -11.7501     3.6982  -3.177  0.00157 **
## Province2Luxembourg       -9.4124     4.1143  -2.288  0.02253 *
## Province2Namur             2.0581     4.3885   0.469  0.63927
## Mean_income_c              7.8208     0.3705  21.108  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 25.37 on 555 degrees of freedom
## Multiple R-squared:  0.7756, Adjusted R-squared:  0.7716
## F-statistic: 191.8 on 10 and 555 DF,  p-value: < 2.2e-16
```

5 Verification: Did the Solution Work?

5.1 VIF After the Fix

```r
car::vif(model_solution) |>
    as.data.frame() |>
    knitr::kable(caption = "GVIF values for the fixed model",
        digits = 4, booktabs = TRUE)
```

Table 5: GVIF values for the fixed model

	GVIF	Df	GVIF^(1/(2*Df))
AdminArea	3.8637	1	1.9656
Province2	5.7855	8	1.1160
Mean_income_c	1.7391	1	1.3187

5.2 Rank of the New Design Matrix

```
X_solution <- model.matrix(~AdminArea + Province2 + Mean_income_c,
    data = df)
cat("Number of columns in X:", ncol(X_solution), "\n")
```

```
## Number of columns in X: 11
```

```
cat("Rank of X:", qr(X_solution)$rank, "\n")
```

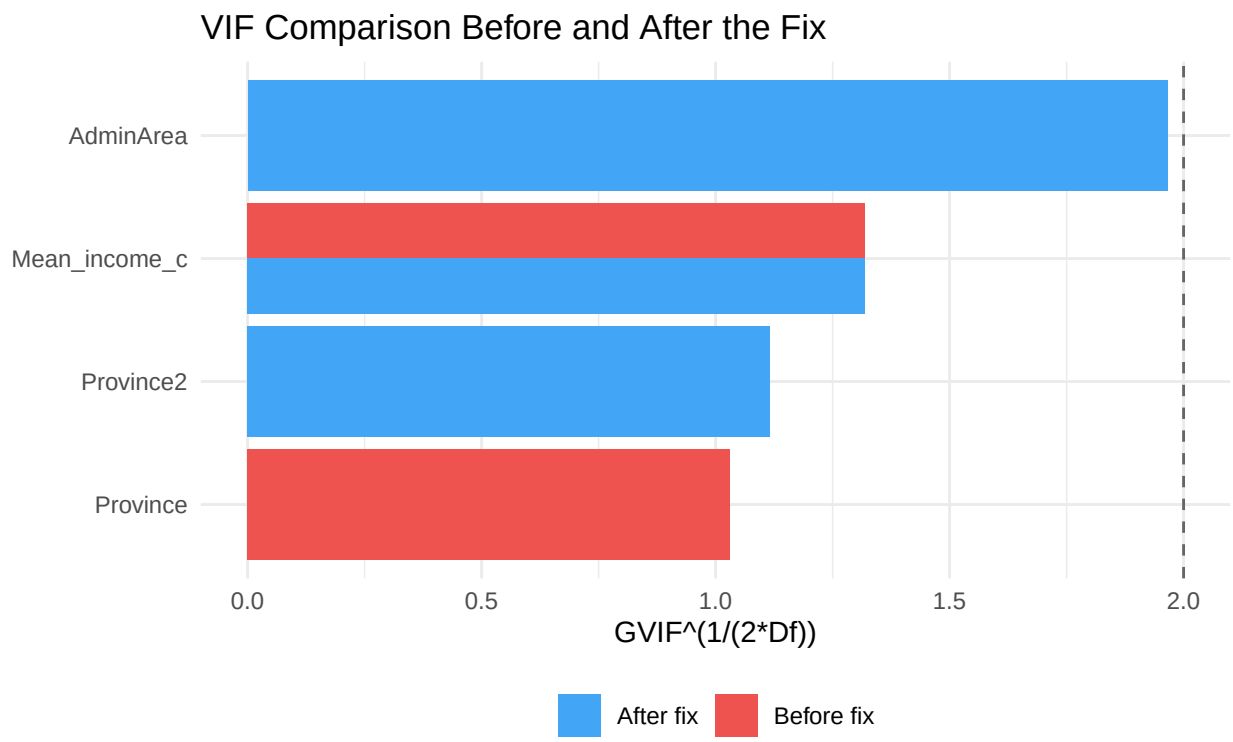
```
## Rank of X: 11
```

```
cat("Rank deficiency:", ncol(X_solution) - qr(X_solution)$rank,
    "\n")
```

```
## Rank deficiency: 0
```

5.3 Visual Comparison of VIF

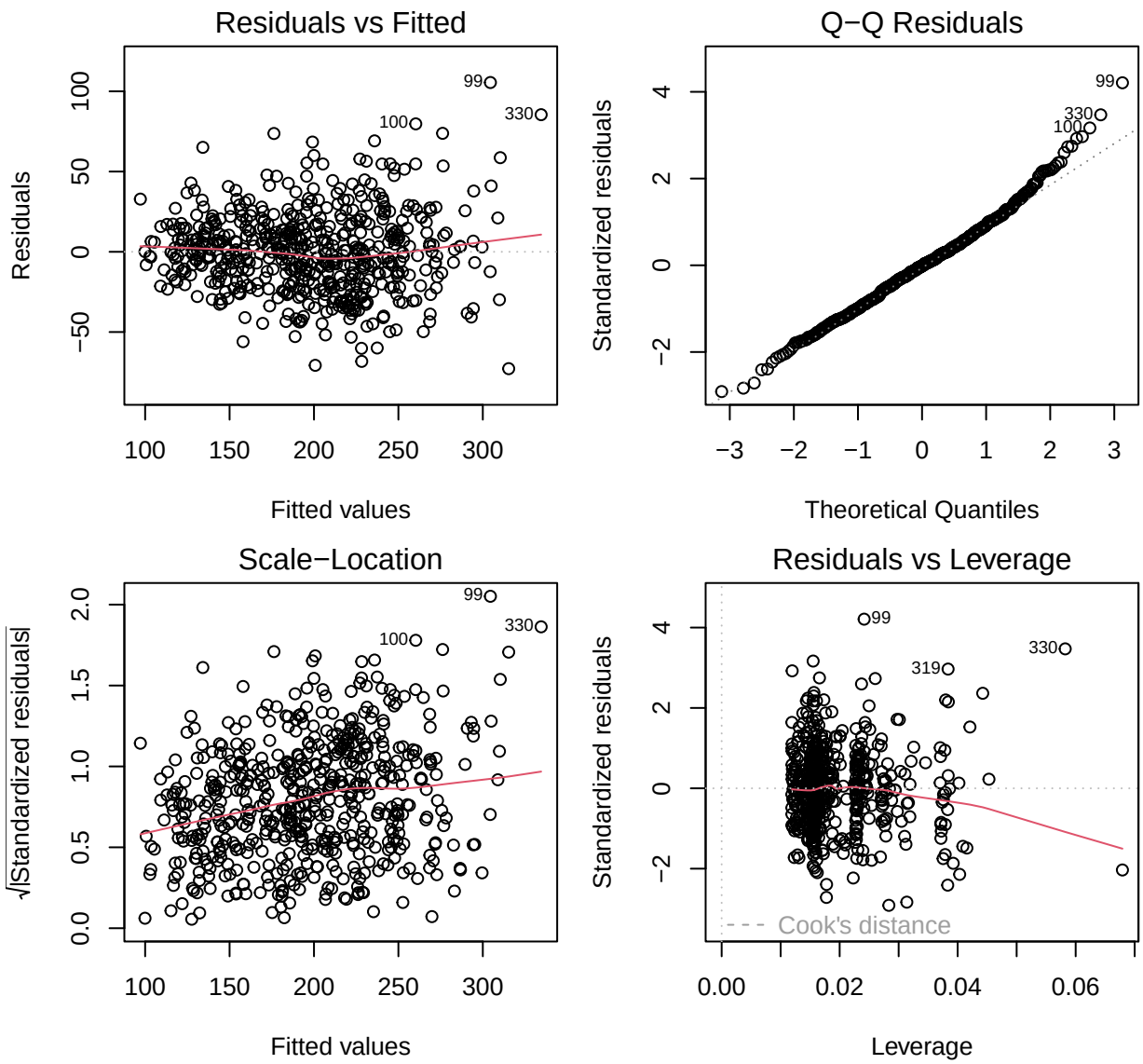
```
model_partial <- lm(med_price_house ~ Province + Mean_income_c,
    data = df)

vif_before <- car::vif(model_partial)
vif_after <- car::vif(model_solution)

vif_before_df <- data.frame(variable = names(vif_before[,
    3]), vif = vif_before[, 3], model = "Before fix")

vif_after_df <- data.frame(variable = names(vif_after[,
    3]), vif = vif_after[, 3], model = "After fix")

vif_combined <- bind_rows(vif_before_df, vif_after_df)

ggplot(vif_combined, aes(x = reorder(variable, vif), y = vif,
    fill = model)) + geom_col(position = "dodge") + geom_hline(yintercept = 2,
    linetype = "dashed", color = "gray40") + coord_flip() +
    scale_fill_manual(values = c(`Before fix` = "#EF5350",
        `After fix` = "#42A5F5")) + labs(title = "VIF Comparison Before and After the Fix",
    x = NULL, y = "GVIF^(1/(2*Df))", fill = NULL) + theme_minimal(base_size = 11) +
    theme(legend.position = "bottom")
```



5.4 Model Diagnostics

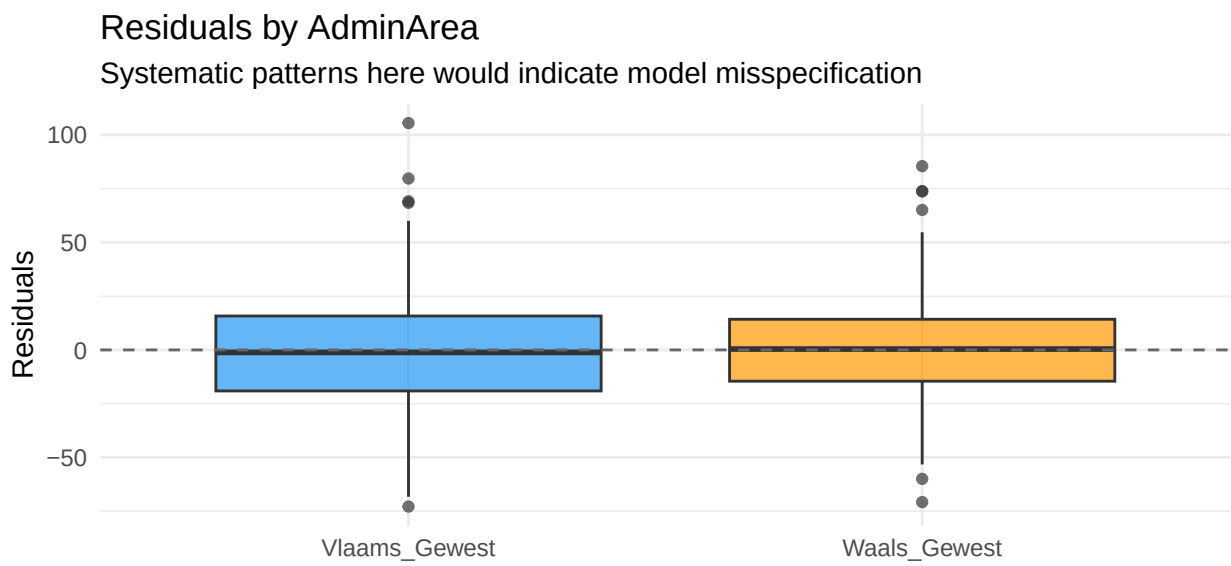
Beyond VIF, we verify the classical regression assumptions on the fitted model. These diagnostics tell us how much we can trust the coefficient estimates and standard errors for inference.

```
par(mfrow = c(2, 2), mar = c(4, 4, 2, 1))
plot(model_solution, which = c(1, 2, 3, 5))
```




```r
par(mfrow = c(1, 1))
```

```r
df$residuals <- residuals(model_solution)

ggplot(df, aes(x = AdminArea, y = residuals, fill = AdminArea)) +
    geom_boxplot(alpha = 0.7) + geom_hline(yintercept = 0,
    linetype = "dashed", color = "gray40") + scale_fill_manual(values = c(Vlaams_Gewest = "#2196F3",
    Waals_Gewest = "#FF9800")) + labs(title = "Residuals by AdminArea",
    subtitle = "Systematic patterns here would indicate model misspecification",
    x = NULL, y = "Residuals") + theme_minimal(base_size = 11) +
    theme(legend.position = "none")
```



If the residual spread differs substantially between regions, this confirms heteroscedasticity. In that case, coefficient estimates remain unbiased but standard errors should be interpreted with caution. For a methodological demonstration like this one, that is acceptable. For formal inference in a production setting, heteroscedasticity-consistent standard errors or a multilevel model would be more appropriate.

6 Interpretation: Reading Coefficients Under Sum-to-Zero Contrasts

6.1 The Intercept

```
cat("Intercept (global mean housing price):", round(coef(model_solution)["(Intercept)"],
    2), "thousands\n")
```

```
## Intercept (global mean housing price): 195.48 thousands
```

6.2 Province Coefficients

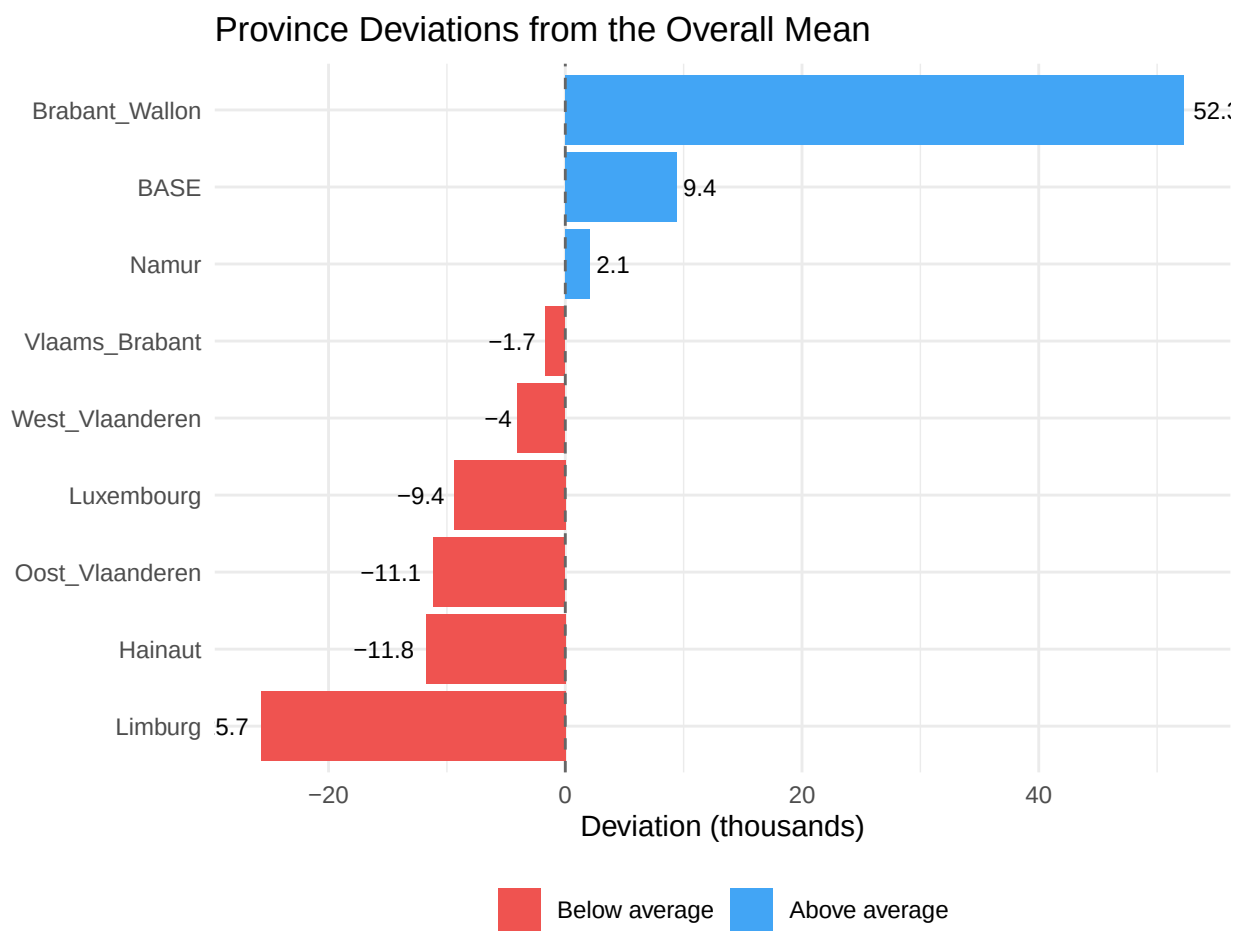
$$\beta_{\text{BASE}} = -\sum_{k \neq \text{BASE}} \beta_k$$

```
province_coefs <- coef(model_solution)[grep("Province2",
    names(coef(model_solution)))]
beta_base <- -sum(province_coefs)
all_province_coefs <- c(province_coefs, BASE = beta_base)

comparison <- data.frame(Province = gsub("Province2", "",
    names(all_province_coefs)), Coefficient = round(all_province_coefs,
    2)) |>
    arrange(desc(Coefficient))

knitr::kable(comparison, caption = "Province deviations from the global mean",
    row.names = FALSE, booktabs = TRUE)
```

Table 6: Province deviations from the global mean

Province	Coefficient
Brabant_Wallon	52.29
BASE	9.40
Namur	2.06
Vlaams_Brabant	-1.68
West_Vlaanderen	-4.05
Luxembourg	-9.41
Oost_Vlaanderen	-11.14
Hainaut	-11.75
Limburg	-25.71

```r
ggplot(comparison, aes(x = reorder(Province, Coefficient),
    y = Coefficient, fill = Coefficient > 0)) + geom_col() +
    geom_hline(yintercept = 0, linetype = "dashed", color = "gray40") +
    geom_text(aes(label = round(Coefficient, 1)), hjust = ifelse(comparison$Coefficient >
        0, -0.2, 1.2), size = 3) + coord_flip() + scale_fill_manual(values = c(`TRUE` = "#42A5F5",
    `FALSE` = "#EF5350"), labels = c(`TRUE` = "Above average",
    `FALSE` = "Below average")) + labs(title = "Province Deviations from the Overall Mean",
    x = NULL, y = "Deviation (thousands)", fill = NULL) +
    theme_minimal(base_size = 11) + theme(legend.position = "bottom")
```



6.3 A Concrete Example

```r
beta_vb <- coef(model_solution)["Province2Vlaams_Brabant"]
beta_ha <- coef(model_solution)["Province2Hainaut"]
diff <- beta_vb - beta_ha

cat("Vlaams_Brabant coefficient:", round(beta_vb, 2), "\n")
```

```
## Vlaams_Brabant coefficient: -1.68
```

```r
cat("Hainaut coefficient:", round(beta_ha, 2), "\n")
```

```
## Hainaut coefficient: -11.75
```

```r
cat("Expected price difference:", round(diff, 2), "thousands\n")
```

```
## Expected price difference: 10.07 thousands
```

A municipality in Vlaams_Brabant is expected to have a housing price approximately that many thousands higher than a comparable municipality in Hainaut, holding mean income constant.

7 Applicability and Limitations

7.1 When This Solution Is Appropriate

This solution is appropriate when the analyst needs both hierarchical levels in the model simultaneously for interpretability, the sample size is large enough for the Central Limit Theorem to protect inference under moderate violations, and the modeling framework is classical linear regression.

7.2 When to Consider Alternatives

If the primary goal is to model within-group correlation correctly, a multilevel model with `lme4` is more appropriate. If heteroscedasticity across groups is severe, heteroscedasticity-consistent standard errors via the `sandwich` package should be used. If the response variable requires transformation for valid inference, the log-transformed response can be used with the understanding that coefficients represent approximate percentage deviations.

7.3 The General Recipe

Table 7: General recipe for solving perfect multicollinearity

Step	Action
1	Identify the two hierarchical levels that cause the problem
2	Check homoscedasticity across groups with Levene's test
3	Choose a BASE category: most frequent province per region
4	Recode the lower level variable replacing chosen categories with BASE
5	Set BASE as the last factor level
6	Apply contr.sum to both variables
7	Verify with VIF and rank analysis that the deficiency is resolved

7.4 Summary

This document demonstrates a methodological solution for perfect multicollinearity caused by nested geographic variables. The solution is statistically valid under classical linear regression assumptions and is especially valuable when interpretability of fixed coefficients at both geographic levels is the primary goal.

It is not a replacement for multilevel models but a practical and accessible alternative when the analyst works within a classical regression framework and needs both geographic levels with directly interpretable coefficients.

This solution was developed during the course Applied Statistical Models I at ESPOL, Ecuador, as part of a group project. The methodological contribution documented here was developed independently by me.